

Foundation of Blockchain Technology (IS501)

Pre-Reading Material – Lecture 7

Types of Blockchain

Estimated Reading Time: 15–20 minutes

Reference Reading

1. Bhutta, M. N. M., et al. (2021). *A Survey on Blockchain Technology: Evolution, Architecture and Security*. IEEE Access.
2. Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.

Why This Topic Matters

Not every blockchain network works in the same way. Blockchain networks can be classified according to **who can participate, who can validate transactions, and who controls the network**, making different types suitable for different applications.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the different types of blockchain networks.
- Differentiate between **public and private blockchains**.
- Explain the concept of **consortium blockchain**.
- Understand **permissioned and permissionless** blockchains.
- Identify suitable applications for different blockchain types.



Activate Your Prior Knowledge

1. Open or Restricted Access

Imagine a system where anyone can join and another system where only approved users can participate.

Think: Which system would be more suitable for a public cryptocurrency and which for a company's internal records?

2. Who Controls the Network?

Suppose several banks want to maintain a shared blockchain.

Think: Would it make sense for every person on the Internet to validate their transactions?

3. Public Information

Some blockchain networks allow anyone to view transactions, while others restrict access.

Think: Why might organizations prefer controlling who can access their blockchain?

Prerequisite Concepts

You should be familiar with:

- Blockchain architecture
- Blockchain nodes
- Distributed ledger
- Consensus mechanisms
- Decentralization
- Proof of Work and Proof of Stake



Core Concepts

1. Types of Blockchain

Blockchain networks are commonly classified as **public, private, and consortium blockchains** based on participation and management. Public networks allow open participation, while private and consortium networks restrict participation.

2. Public Blockchain

A public blockchain is open to anyone who wants to join the network. Participants can generally read transactions, submit transactions, and participate in maintaining the shared ledger.

3. Permissionless Blockchain

A permissionless blockchain does not require prior approval to join the network. Participants can interact with the blockchain and may participate in the consensus process according to the protocol rules.

4. Private Blockchain

A private blockchain restricts participation to authorized users, generally within a single organization. Participants are invited and may be assigned specific roles for maintaining the blockchain.

5. Permissioned Blockchain

A permissioned blockchain requires authorization before participants can join or perform specific activities. Participant identities are generally known, making it suitable for controlled organizational environments.



6. Consortium Blockchain

A consortium blockchain is operated by multiple organizations rather than a single organization. Participation is restricted to invited and trusted entities from the participating organizations.

7. Public vs Private Blockchain

Public blockchains provide open participation and greater decentralization, while private blockchains restrict participation to authorized entities. Private networks can therefore provide greater control and efficiency in organizational environments.

8. Permissioned vs Permissionless

Permissionless networks use open participation and commonly rely on economic or cryptographic mechanisms for trust. Permissioned networks emphasize controlled access, known identities, and organizational governance.

9. Decentralization

Decentralization means that blockchain information and control are distributed across multiple nodes rather than being maintained by a single central authority. The degree of decentralization varies among different blockchain types.

10. Choosing the Right Blockchain

The choice of blockchain type depends on factors such as participation, privacy, governance, transparency, and application requirements. Public blockchains suit open ecosystems, while private and consortium networks suit controlled organizational environments.

Guided Reading Questions

1. What are the major types of blockchain?
2. What is the difference between public and private blockchain?
3. What does permissionless mean?
4. What does permissioned mean?
5. How does a consortium blockchain differ from a private blockchain?
6. Why are public blockchains generally more decentralized?
7. Why might an organization choose a permissioned blockchain?
8. Which type of blockchain would be suitable for a group of banks sharing information?

Reflection Activity

Write 100–150 words explaining which type of blockchain would be most appropriate for a **banking consortium**. Consider participation, privacy, governance, decentralization, and security in your answer.

Self-Assessment

1. A blockchain that allows anyone to participate is generally called a _____ blockchain.
2. A blockchain that requires authorization to participate is called a _____ blockchain.
3. Bitcoin is an example of a _____ blockchain.
4. A private blockchain is generally controlled by:
 - A. Anyone on the Internet
 - B. A single organization
 - C. No organization
 - D. Random users
5. A consortium blockchain is operated by:



- A. A single individual
 - B. Multiple organizations
 - C. Only cryptocurrency miners
 - D. Government only
6. Public blockchains generally provide _____ participation.
7. Permissioned blockchains generally have _____ participant identities.
8. Which type is most suitable when several organizations want to jointly maintain a blockchain?
- A. Public
 - B. Private
 - C. Consortium
 - D. Personal
9. Public blockchains generally provide greater _____ than private blockchains.
10. The primary difference between permissioned and permissionless blockchains is _____.

Key Takeaway

Blockchain networks differ mainly in **who can join, who can participate in consensus, and who controls the network**. Public/permissionless blockchains emphasize open participation and decentralization, while private/permissioned and consortium blockchains provide greater control over participation and governance.

